

MSDM5004 Spring 2026
Homework 1 Solution

(Non-exact results are expressed in either the long or the short format of MATLAB.)

1. Finding roots

- (1) The iteration algorithm of Newton's method is

$$\begin{aligned}x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)} \\&= x_n - \frac{1 - 4(e^{x_n} - 1) \cos x_n + 2x_n^2 + \cos 2x_n}{-4e^{x_n} \cos x_n + 4(e^{x_n} - 1) \sin x_n + 4x_n - 2 \sin 2x_n}.\end{aligned}$$

Starting from $x_0 = 0.6$,

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0.425616990959189 ,$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 0.445287379211599 ,$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 0.445415149643081 ,$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 0.445415155622615 .$$

- (2) (i) For Newton's method, suppose the starting point is $x_0 = 0.6$. It can be implemented in MATLAB as

```
f = @(x) 1 - 4*(exp(x)-1)*cos(x) + 2*x^2 + cos(2*x);
Df = @(x) -4*exp(x)*cos(x) + 4*(exp(x)-1)*sin(x) + 4*x - 2*sin(2*x);
x0 = 100; % just chosen to start the while loop
x1 = 0.6; % the true starting point
while abs(x1-x0) > 1e-5
    x0 = x1;
    x1 = x0 - f(x0) / Df(x0);
end
x1 % 0.445415155622615
```

(ii) For the secant method, suppose the two starting points are $x_0 = 0$ and $x_1 = 0.6$. It can be implemented in MATLAB as

```
f = @(x) 1 - 4*(exp(x)-1)*cos(x) + 2*x^2 + cos(2*x);
x0 = 0;
x1 = 0.6;
while abs(x1-x0) > 1e-5
    x2 = x1 - f(x1)*(x1-x0) / ( f(x1)-f(x0) );
    x0 = x1;
    x1 = x2;
end
x2 % 0.445415156911139
```

2. Interpolation

(1) The Lagrange interpolating polynomial is

$$\begin{aligned} P(x) &= 3 \cdot \frac{(x-0)(x-3)}{(-1-0)(-1-3)} + 8 \cdot \frac{[x-(-1)](x-3)}{[0-(-1)](0-3)} + 9 \cdot \frac{[x-(-1)](x-0)}{[3-(-1)](3-0)} \\ &= \frac{3}{4}(x^2 - 3x) - \frac{8}{3}(x^2 - 2x - 3) + \frac{9}{12}(x^2 + x) \\ &= -\frac{7}{6}x^2 + \frac{23}{6}x + 8. \end{aligned}$$

(2) Therefore, $f(2)$ is approximated as $P(2) = 11$.

3. Least squares fitting

Let

$$A = \begin{pmatrix} 3.4212 & 1 \\ 3.9279 & 1 \\ 5.1779 & 1 \\ 6.8251 & 1 \\ 11.8603 & 1 \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} 13.5339 \\ 15.1001 \\ 18.9670 \\ 23.6083 \\ 36.9070 \end{pmatrix}.$$

The least squares polynomial of degree 1 has the form $y = mx + c$, and its parameters can be obtained by solving $\begin{pmatrix} m \\ c \end{pmatrix} = (A^T A)^{-1}(A^T b)$.

$$\begin{aligned}
\begin{pmatrix} m \\ c \end{pmatrix} &= (A^\top A)^{-1} (A^\top b) = \begin{pmatrix} 241.1924 & 31.2124 \\ 31.2124 & 5 \end{pmatrix}^{-1} (A^\top b) \\
&= \begin{pmatrix} 0.0216 & -0.1347 \\ -0.1347 & 1.0408 \end{pmatrix} \begin{pmatrix} 802.6802 \\ 108.1163 \end{pmatrix} \\
&= \begin{pmatrix} 2.756580574334862 \\ 4.415360896326117 \end{pmatrix}
\end{aligned}$$

Therefore, the least squares error is

$$\sqrt{\sum_{i=1}^5 (mx_i + c - y_i)^2} = 0.616384919692731 .$$

4. Singular value decomposition

The least squares polynomial of degree 2 has the form $y = a_0 + a_1x + a_2x^2$. Let

$$a = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix}, \quad b = \begin{pmatrix} 1.8547 \\ 1.9347 \\ 2.1013 \\ 2.2613 \\ 2.5880 \\ 2.7480 \end{pmatrix}, \quad \text{and} \quad A = \begin{pmatrix} 1 & 1.0700 & 1.0700^2 \\ 1 & 1.2270 & 1.2270^2 \\ 1 & 1.5410 & 1.5410^2 \\ 1 & 1.8550 & 1.8550^2 \\ 1 & 2.4830 & 2.4830^2 \\ 1 & 2.7970 & 2.7970^2 \end{pmatrix}.$$

Perform SVD on A to get

$$\begin{aligned}
U &= \begin{pmatrix} -0.1345 & 0.5443 & 0.5808 & 0.0391 & -0.1875 & -0.5582 \\ -0.1665 & 0.5144 & 0.2102 & 0.1878 & 0.5186 & 0.5994 \\ -0.2417 & 0.4207 & -0.3191 & -0.7119 & -0.3400 & 0.2009 \\ -0.3317 & 0.2817 & -0.5658 & 0.6367 & -0.2752 & -0.0970 \\ -0.5560 & -0.1319 & -0.2113 & -0.2163 & 0.6238 & -0.4392 \\ -0.6904 & -0.4065 & 0.3899 & 0.0646 & -0.3397 & 0.2941 \end{pmatrix}, \\
\Sigma &= \begin{pmatrix} 12.0189 & 0 & 0 \\ 0 & 1.6539 & 0 \\ 0 & 0 & 0.1303 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \text{and} \quad V = \begin{pmatrix} -0.1755 & 0.7394 & 0.6500 \\ -0.3847 & 0.5563 & -0.7366 \\ -0.9062 & -0.3793 & 0.1868 \end{pmatrix}.
\end{aligned}$$

Then, compute

$$c = U^\top b = \begin{pmatrix} -5.1656 \\ 2.0675 \\ 0.0585 \\ -0.0025 \\ -0.0003 \\ -0.0012 \end{pmatrix} \quad \text{and} \quad z = \begin{pmatrix} c_1/s_1 \\ c_2/s_2 \\ c_3/s_3 \end{pmatrix} = \begin{pmatrix} -0.4275 \\ 1.2501 \\ 0.4493 \end{pmatrix}.$$

where $s_i = \Sigma_{ii}$ is the i th singular value of A , so that the parameters of the polynomial can be obtained by solving $a = Vz$.

$$\begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = Vz = \begin{pmatrix} 1.291362376844743 \\ 0.528936422599016 \\ -0.002858814314509 \end{pmatrix}$$

In addition, the least squares error is

$$\sqrt{\sum_{i=1}^5 (a_0 + a_1 x_i + a_2 x_i^2 - y_i)^2} \equiv \sqrt{c_4^2 + c_5^2 + c_6^2} = 0.002818210176010 .$$

The above computation can be implemented in MATLAB as

```
deg = 2; % degree of the polynomial
x = [1.0700 1.2270 1.5410 1.8550 2.4830 2.7970]';
b = [1.8547 1.9347 2.1013 2.2613 2.5880 2.7480]';
A = [ones(size(x))];
for i=1:deg
    A = [A x.^i]; % stack column by column
end

[U S V] = svd(A);
c = U'*b;
z = zeros(deg+1, 1);
for i=1:deg+1
    z(i) = c(i)/S(i, i);
end
a = V*z % [1.291362376844743 0.528936422599016 -0.002858814314509] '
err = sqrt(sum(c(deg+2:end).^2)) % 0.002818210176010
```

Remarks on MATLAB

- If a declarative statement is not ended by a semicolon (;), the result of the statement is printed on the command window automatically.
- A dot (.) means “elementwise” when it precedes another operator, so, for example, “x.^i” raises all elements of x to the i th power.